

Lab Report
Summer 2025

Course Code: **EEE203L**
Course Title: **Electronic Circuits II Laboratory**

Lab Section: **02**

Experiment No.: 3

Name of the experiment: Verification of KCL in AC Circuits

Date of Submission: 09/08/2025

Group No: 3

Group Member Details:

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Objective

The purpose of this experiment is to verify Kirchhoff's Current Law (KCL) for an AC circuit containing resistive, inductive and capacitive elements. By constructing an RLC circuit with series-parallel network and measuring the branch current in phasor form, and confirming that the phasor sum equals the total current, we will verify KCL.

Theoretical Background

KCL states that the algebraic sum of currents at any node is zero: $\sum I = 0$

In AC circuits, voltages and currents are represented as phasors to include both magnitude and phase

Thus KCL in phasor form is: $\vec{I}_1 + \vec{I}_2 + \vec{I}_3 = 0$

Where, $\vec{I}_k = I_k \angle \theta_k$

In AC analysis:

- * Resistor (R): Current in phase with voltage.

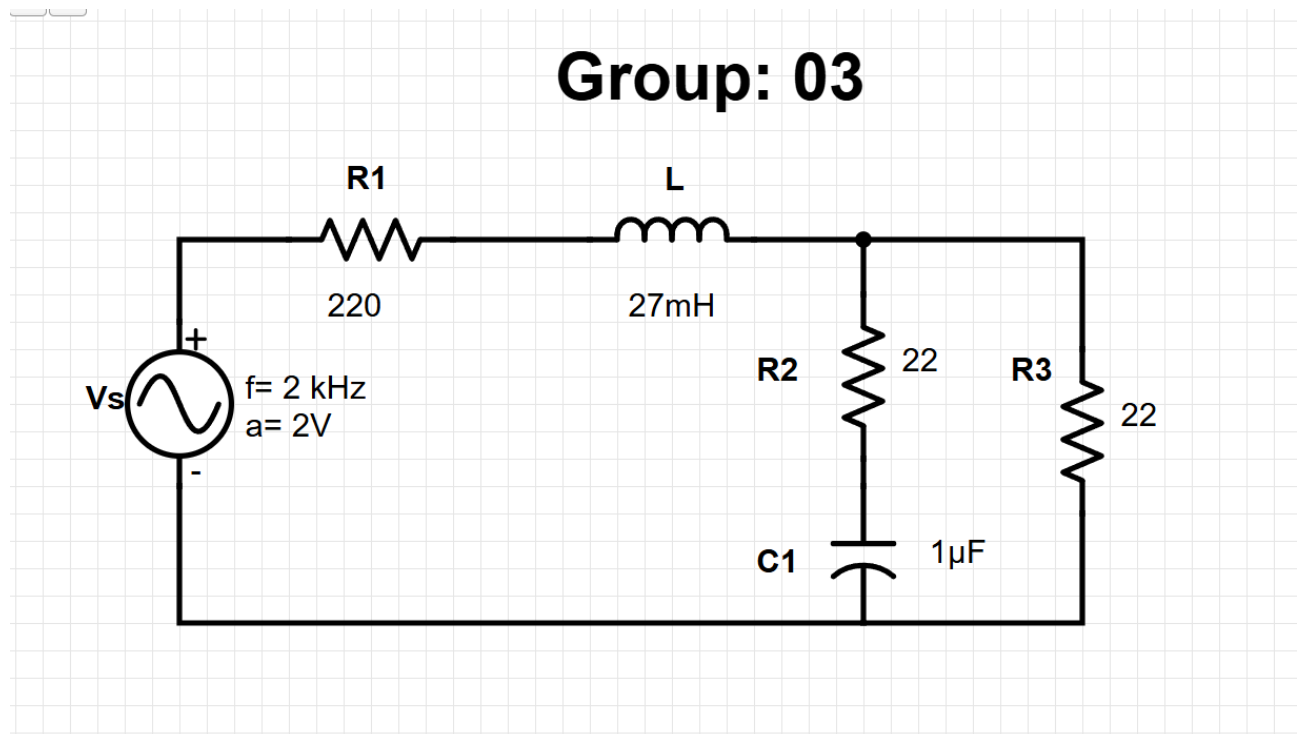
- * Inductor (L): Current lags voltage by 90° .

- * Capacitor (C): Current leads voltage by 90° .

Since, AC branch currents differ both in magnitude and phase, Verifying KCL requires vector (phasor) addition.

Compilation of result and Calculations (with ckt diagram and verification of result):

Circuit Diagram:



Data Table:

Table for I_{R1}								
R_1 (Ω) (using multimeter)	V_{R1a} (V) (Peak Value from Oscilloscope)	$ V_{R1} $ = $V_{R1a}/\sqrt{2}$ (V)	$ I_{R1} $ = $ V_{R1} /R_1$ (mA)	Sign (+/-)	Δt (ms) (From oscilloscope)	f(kHz) (From oscilloscope)	360f Δt (o)	$\angle I_{R1}$ (o)
219	1.75	1.237	5.65	(-)	0.0116	1.99	6.31	$\angle -8.31$

Table for I_{R2}								
R_2 (Ω) (using multimeter)	V_{R2a} (V) (Peak Value from Oscilloscope)	$ V_{R2} $ = $V_{R2a}/\sqrt{2}$ (V)	$ I_{R2} $ = $ V_{R2} /R_2$ (mA)	Sign (+/-)	Δt (ms) (From oscilloscope)	f(kHz) (From oscilloscope)	360f Δt (o)	$\angle I_{R2}$ (o)
22.6	0.036	0.025	1.1	(+)	0.070	1.99	50.15	$\angle +50.15$

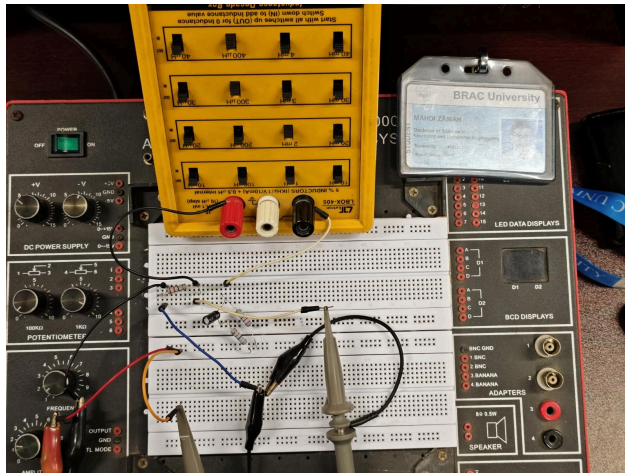
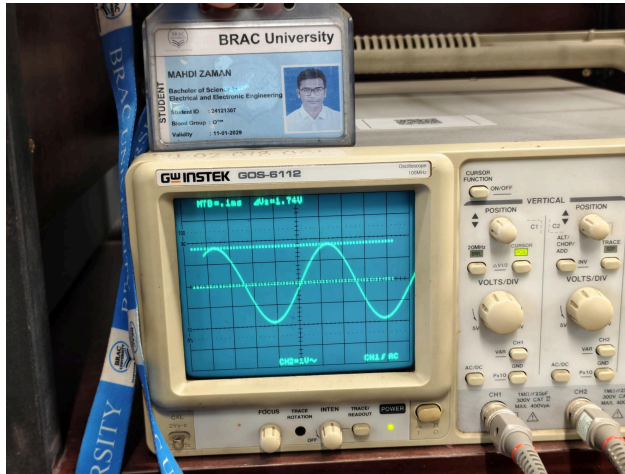
Table for I_{R3}								
R_3 (Ω) (using multimeter)	V_{R3a} (V) (Peak Value from Oscilloscope)	$ V_{R3} $ = $V_{R3a}/\sqrt{2}$ (V)	$ I_{R3} $ = $ V_{R3} /R_3$ (mA)	Sign (+/-)	Δt (ms) (From oscilloscope)	f(kHz) (From oscilloscope)	360f Δt (o)	$\angle I_{R3}$ (o)
22.7	0.160	0.113	4.98	(-)	0.031	1.99	22.208	$\angle -22.208$

Error

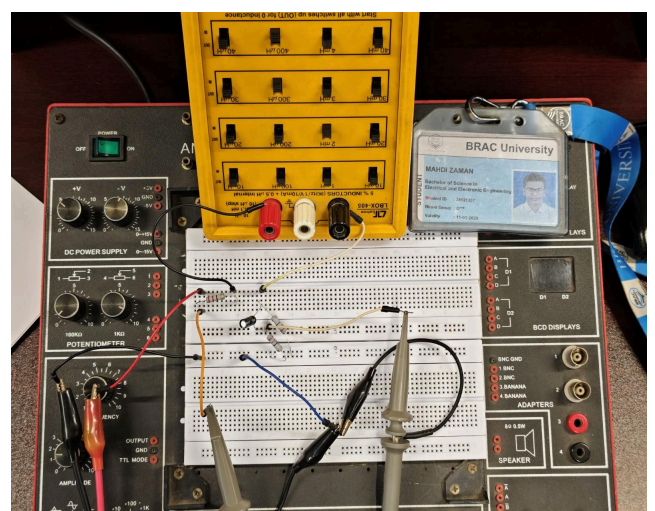
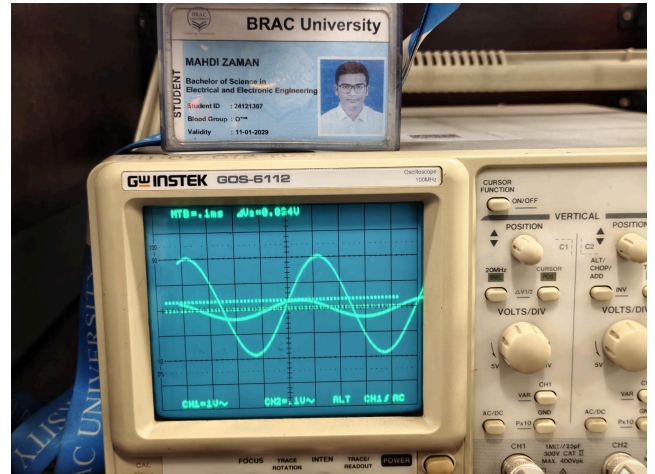
$$\begin{aligned} & [\{ (5.65\angle - 8.31) - (5.42\angle - 11.04) \} / (5.65\angle - 8.31)] \times 100 \\ & = \mathbf{6.19\angle 47.55 \text{ percent}} \end{aligned}$$

Classwork Proof :

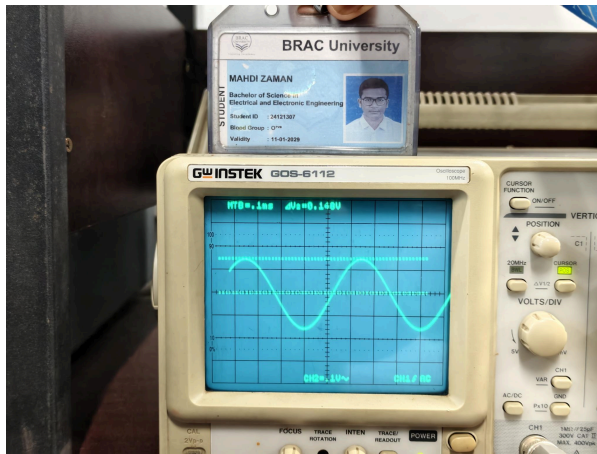
R₁



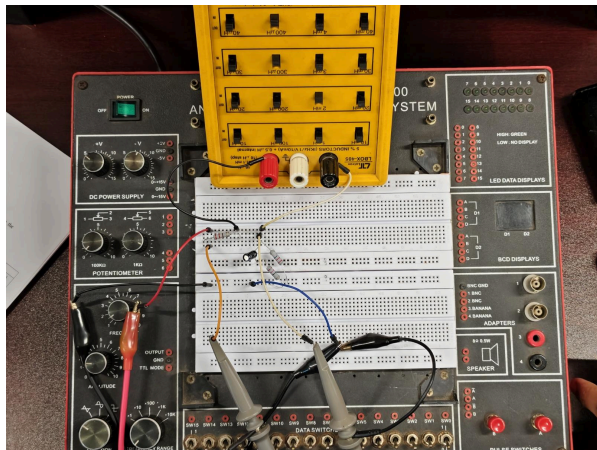
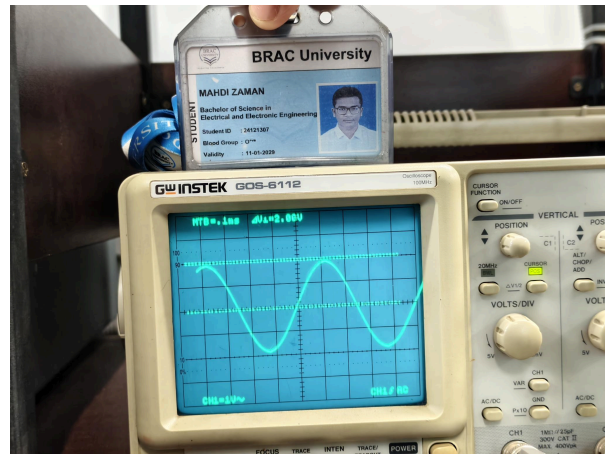
R₂



R_3



V_s



Appendix:

Brac University
Department of Electrical & Electronic Engineering (EEE)
EEE203L – Electrical Circuits II Laboratory

Experiment 3

Name of the Experiment:

Verification of KCL in AC Circuits

Objective:

The objective of this experiment is to study RLC series parallel circuit when energized by an ac source. KCL in phasor form will also be verified.

Circuit Diagram:

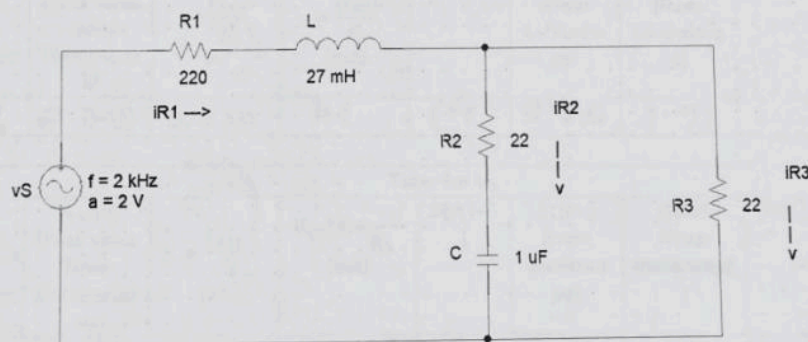


Figure 1 : KCL Circuit

Equipments:

1. Multimeter
2. AC ammeter
3. Resistances – 22Ω (1 piece) and $2.5k\Omega$ (3 pieces)
4. Capacitor – $1\mu F$ (1 piece)
5. Inductor – 2.7 mH (1 piece)
6. Bread Board
7. Wires for connection

Procedure:

- Construct the circuit of figure 1.
- Observe the wave shapes for the currents of I_{R1} , I_{R2} , and I_{R3} and fill up the required table.

Data:

Table for I_{R1}								
R_1 (Ω) (using multimeter)	V_{R1a} (V) (Peak value from Oscilloscope)	$ V_{R1} $ $= \frac{V_{R1a}}{\sqrt{2}}$ (V)	$ I_{R1} $ $= \frac{ V_{R1} }{R_1}$ (mA)	Sign (+/-)	Δt (ms) (From oscilloscope)	f(kHz) (From oscilloscope)	$360f\Delta t$ (o)	$\angle I_{R1}$ (o)
219	1.75	1.237	5.65	(-)	0.0116	1.99	8.31	$\angle -8.31$

Table for I_{R2}								
R_2 (Ω) (using multimeter)	V_{R2a} (V) (Peak value from Oscilloscope)	$ V_{R2} $ $= \frac{V_{R2a}}{\sqrt{2}}$ (V)	$ I_{R2} $ $= \frac{ V_{R2} }{R_2}$ (mA)	Sign (+/-)	Δt (ms) (From oscilloscope)	f(kHz) (From oscilloscope)	$360f\Delta t$ (o)	$\angle I_{R2}$ (o)
22.6	0.036	0.025	1.1	(+)	0.070	1.99	50.15	$\angle 50.15$

Table for I_{R3}								
R_3 (Ω) (using multimeter)	V_{R3a} (V) (Peak value from Oscilloscope)	$ V_{R3} $ $= \frac{V_{R3a}}{\sqrt{2}}$ (V)	$ I_{R3} = \frac{ V_{R3} }{R_3}$ (mA)	Sign (+/-)	Δt (ms) (From oscilloscope)	f(kHz) (From oscilloscope)	$360f\Delta t$ (o)	$\angle I_{R3}$ (o)
22.7	0.160	0.113	4.98	(-)	0.031	1.99	22.208	$\angle -22.208$

Table for $I_{R2} + I_{R3}$ $\rightarrow 5.42\angle -11.04$	
$ I_{R2} + I_{R3} $ (mA)	$\angle I_{R2} + I_{R3}$ (o)
5.42	-11.04

$$\text{error} = \frac{5.65\angle -8.31 - 5.42\angle -11.04}{5.65\angle -8.31} \times 100\%$$

$$= 6.29\angle 47.58\%$$

Conclusion:

We built a circuit where one resistor was in one branch and two other resistors were in parallel in another branch. Then we sent AC (wave-like electricity) through it and measured the currents in each branch. We added the currents from the parallel branch (with their wave angles) and compared the total to the current in the single-branch path. We found that:

1. The total we calculated was close to the measured value.
2. There was some error **6.19%47.55 %**, likely due to measurement inaccuracies or because real components are not ideal.

So, even though there were small differences, the experiment mostly matched what KCL says should happen. This confirms that KCL works in AC circuits just like in DC ones, but phase angles must be handled carefully.